

Design and Analysis of Microwave Antenna Sensors for Early Breast Tumor Detection

Pre-Project Mid-Term Progress Evaluation

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September 15, 2026

1. Introduction & Problem Statement

Clinical Context & Motivation

- Breast cancer is the leading cause of oncological mortality among women worldwide.
- Early screening significantly enhances the 5-year relative survival rate to $> 95\%$.

Limitations of Conventional Modalities

- **X-ray Mammography:** Ionizing radiation hazard, physical compression pain, low sensitivity ($\sim 48\%$) in dense fibroglandular breasts.
- **MRI / Ultrasound:** High operational expense, poor operator portability, and variable resolution.

Microwave Imaging (MWI) Solution

- **Non-ionizing** low-power electromagnetic radiation ($< 1 \text{ mW}$).
- **Cost-effective**, lightweight, and comfortable patient setup.
- Exploits inherent **dielectric contrast** between normal and malignant breast tissues.

Target: Safe, non-invasive, pre-clinical screening.

2. Working Principle & Biological Basis

Dielectric Contrast Mechanism

- Malignant tumors possess higher vascularity, sodium concentration, and bound water content.
- Yields a substantial contrast in permittivity (ϵ_r) and conductivity (σ) (2 : 1 to 5 : 1).

Electromagnetic Boundary Reflection

Reflection coefficient at tissue interface:

$$\Gamma = \frac{\sqrt{\epsilon_{r,tumor}^*} - \sqrt{\epsilon_{r,normal}^*}}{\sqrt{\epsilon_{r,tumor}^*} + \sqrt{\epsilon_{r,normal}^*}}$$

$$\text{where } \epsilon_r^* = \epsilon_r - j \frac{\sigma}{\omega \epsilon_0}$$

Tissue Dielectric Properties (at 3–5 GHz)

Tissue Type	Permittivity (ϵ_r)	Cond. (σ , S/m)
Skin (Dry/Wet)	35 – 38	1.2 – 1.6
Adipose (Fat)	3.5 – 5.5	0.05 – 0.15
Fibroglandular	12 – 20	0.4 – 0.9
Malignant Tumor	50 – 60	3.0 – 4.5

⇒ **Observation:** Substantial back-scattered microwave reflection occurs when pulses interact with high-contrast tumor boundaries.

3. Project Objectives & Scope

Primary Project Objectives

- ➊ **Antenna Design:** Design an ultra-wideband (UWB) compact microstrip patch antenna operating within 2.5 GHz – 8.0 GHz band.
- ➋ **High Gain & Directivity:** Optimize antenna layout to achieve directional radiation pattern, low return loss ($S_{11} < -10$ dB), and adequate tissue penetration.
- ➌ **Tissue Phantom Modeling:** Model realistic heterogeneous breast tissue phantoms (skin, adipose, glandular, and spherical tumors from 3 mm to 10 mm).
- ➍ **Scattering Parameter Analysis:** Characterize differential reflection signatures (S_{11} , S_{21}) in the presence vs. absence of tumor.

In-Scope

- CST Studio.
- Synthetic dielectric breast phantom recipe.
- Hardware antenna prototype fabrication.

Out-of-Scope

- In-vivo human clinical trials.
- High-cost full 3D confocal robotic gantry.

4. Literature Survey Summary

Author & Year	Antenna Geometry		Bandwidth / Features	Observed Limitations
Elsheakh et al. (2021)	Antipodal	Vivaldi Antenna	2.4–9.0 GHz, high end-fire gain	Bulky planar dimensions; complex array integration.
Rahman et al. (2022)	Slotted Patch	Microstrip	3.1 – 7.5 GHz, compact profile	Lower penetration depth in dense breast phantoms.
Mahmud et al. (2023)	Flexible Monopole with EBG		2.8 – 6.2 GHz, SAR compliant	Difficult phantom positioning; higher fabrication cost.
Our Proposed Approach	Defected Structure (DGS)	Ground Microstrip Patch	3.0 – 6.5 GHz, compact ($30 \times 32 \text{ mm}^2$), directional	Optimized balance between size, bandwidth, and tissue coupling.

5. Proposed Methodology & Execution Pipeline

1. Antenna Design

Substrate selection, microstrip feeding, and slot geometry.

2. EM Simulation

CST Studio optimization for $S_{11} < -10$ dB.

3. Phantom Setup

Numerical heterogeneous breast tissue modeling.

4. Detection Scheme

Differential S_{11} backscatter contrast extraction.

Planned Simulation Domain

- **Substrate Choice:** Low-loss substrate evaluation (FR4 vs. Rogers RT5880).
- **Antenna Form Factor:** Compact planar microstrip patch with Defected Ground Structure (DGS).
- **Phantom Model:** Multi-layer breast model (Skin, Adipose, Fibroglandular, Tumor).

Detection Principle to Implement

- Radiate low-power UWB microwave signal into the phantom.
- Record reference reflections from normal breast model ($S_{11,\text{ref}}$).
- Record perturbations introduced by tumor presence ($S_{11,\text{tumor}}$).
- Compute differential response:

$$\Delta S_{11} = |S_{11,\text{tumor}} - S_{11,\text{ref}}|$$

6. Mid-Term Progress: Literature Insights & Design Targets

Target Design Specifications (From Survey)

Parameter	Proposed Value	Target
Target Frequency Band	3.1 GHz – 6.0 GHz (UWB)	
Candidate Substrate	Rogers RT5880 / FR4	
Target Dielectric (ϵ_r)	2.2 – 4.4	
Substrate Thickness (h)	1.57 – 1.60 mm	
Impedance Matching	50 Ω Microstrip Line	
Return Loss Goal	$S_{11} \leq -10$ dB across band	
Antenna Dimensions	Compact 30×32 mm ²)	($\approx$)

Progress Completed Till Mid-Term

- Conducted extensive literature review on microwave breast imaging modalities.
- Compared patch, monopole, and Vivaldi topologies to identify optimal tradeoffs.
- Established mathematical formulas for patch sizing and edge feeding.
- Shortlisted dielectric property values for healthy tissue vs. malignant tumors.

Mid-Term Milestone

Theoretical groundwork, parameter selection, and tool setup finalized.

7. Expected Outcomes & Post-Mid-Term Simulation Plan

Planned Computational Setup

- Build and tune CAD microstrip antenna model in CST Microwave Studio.
- Sweep slot dimensions to achieve broadband impedance matching.
- Verify far-field radiation patterns and directional front-to-back ratio.

Expected Antenna Performance

- Return loss $S_{11} < -15$ dB at center resonant frequency.
- Peak gain ≥ 3.5 dBi directed into target area.
- Specific Absorption Rate (SAR) within regulatory safety limits (< 1.6 W/kg).

Expected Detection Behavior

- Significant change in backscattered pulse magnitude due to the high dielectric contrast ($\approx 5 : 1$).
- Detectable phase/frequency shifts in S_{11} for tumors of diameter ≥ 4 mm.

Immediate Next Action

Begin 3D model creation and port excitation in EM software immediately after mid-term evaluation.

8. Project Roadmap & Post-Mid-Term Timeline

Project Execution Schedule (Months 1–7)

Task / Milestone	M1-2	M2-3	M3-5	M5-7
Literature Review & Parameter Selection	Done	–	–	–
Antenna Modeling & CST/HFSS Optimization	Active	Active	–	–
Multi-Layer Breast Phantom Simulation	–	Planned	Active	–
Differential S_{11} Backscatter Analysis	–	–	Planned	–
Prototype PCB Fabrication & VNA Testing	–	–	Planned	Planned
Final Documentation & Project Defense	–	–	–	Final

Phase 1: Simulation & Tuning

- CST/HFSS port excitation setup.
- Optimize patch & DGS for $S_{11} < -10$ dB.
- Evaluate SAR regulatory thresholds.

Phase 2 & 3: Analysis & Testing

- Insert heterogeneous tumor models.
- Extract baseline vs. tumor S_{11} contrast.
- Fabricate prototype for VNA validation.

9. Key References & Discussion

Key References

- ① E. C. Fear, S. C. Hagness, P. M. Meaney, et al., "Enhancing breast tumor detection with near-field microwave imaging," *IEEE Microwave Magazine*, vol. 3, no. 1, pp. 48–56, 2002.
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- ③ S. K. Kakarla and S. K. Patnaik, "Ultra-wideband compact microstrip antenna with DGS for biomedical microwave imaging," *IEEE Trans. Antennas Propag.*, vol. 69, pp. 7114–7122, 2021.
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Thank You!

Questions & Feedback Welcome